Chapter 1. Statistical Learning, Data and Exploration

Lectures 1–3 (overview of statistics/ML/AI, statistical learning, data as probabilistic evidence, assumptions, data structures, EDA)

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1 Statistics, Machine Learning, AI and Data Mining

1.1 Where are we? A statistician's view

- **Statistics**: modeling the data-generating process under uncertainty.
- **Statistical learning**: estimating $P(Y \mid X)$ from limited samples.
- **Machine learning**: optimization for prediction.
- **Data mining**: discovery of patterns and structure in complex data.

AI $\supset$ ML $\supset$ deep learning; statistics underpins many ML methods and models uncertainty.

Example: one problem, four perspectives — customer churn

AI: decide retention strategies. **ML**: learn $\hat{f}(X)$ for the churn probability. **Statistics**: which variables matter, and how certain are we? **Data mining**: hidden segments and behavioral structure.

Data are *traces of a hidden generative process*; learning tries to understand that process.

2 What Is Statistical Learning?

Statistical learning estimates unknown probabilistic relationships from finite data.

- **Supervised learning**: estimate $P(Y \mid X)$ from labeled data.
- **Unsupervised learning**: uncover structure in $P(X)$ without labels.

Definition: notation

Outcome (response, target) Y; predictor vector $X = (X_1, \ldots, X_p)$; data $\{(X_i, Y_i)\}_{i=1}^n$. **Regression** if Y is quantitative, **classification** if Y is categorical.

Supervised learning: model $P(Y \mid X)$	Unsupervised learning: model $P(X)$
prediction: perform well on unseen data **inference**: understand how X influences Y **uncertainty**: reliability of predictions and estimates	**structure estimation**: properties of $P(X)$ **clustering**: latent groups, multimodality **representation learning**: low-dimensional summaries of variability

Learning is *generalization under uncertainty.*

2.1 Applications

Medical risk prediction, spam detection, handwritten-digit recognition, income versus demographics.

Example: spam data

4601 e-mails to "George" (HP Labs), labeled spam or e-mail; features: relative frequencies of 57 common words and punctuation marks. Largest spam/e-mail differences: `george` (0.00 vs 1.27%), `you` (2.26 vs 1.27), `hp` (0.02 vs 0.90), `free` (0.52 vs 0.07), `!` (0.51 vs 0.11), `edu` (0.01

vs 0.29), `remove` (0.28 vs 0.01).

3 Data as Probabilistic Evidence

3.1 Sample versus population

Observations are realizations of random variables from an unknown data-generating process: **stochastic evidence, not absolute truth**.

Definition: the inferential setting

Observed sample $\mathcal{D}_n = \{Z_1, \ldots, Z_n\}$, population distribution $Z \sim P_0$, target $\theta(P_0)$ (a center, a dispersion, a regression coefficient, ...).

Sampling variability. Repeating n coin tosses gives a different $\hat{p} = \bar{Z}$ each time, $\mathrm{Var}(\hat{p}) = p(1-p)/n$.

3.2 Assumptions and identifiability

Data alone do not determine the target, $\mathcal{D}_n \not\Rightarrow \theta(P_0)$; we restrict $P_0 \in \mathcal{P}_{\text{model}}$, within which identification becomes possible.

Example: five rainy days

Rain, Rain, Rain, Rain, Rain. Does it always rain, did the rainy season start, or were we unlucky? Only an assumption (i.i.d. days, a seasonal model) picks one explanation.

Definition: data-generating process

$$Y = f(X) + \varepsilon,$$

where f is the underlying mechanism and ε captures randomness and unobserved factors.

A model specifies a **class** for f, which fixes flexibility and tools:

Function class	Example	Tools
linear models	$f(x) = x^\top \beta$	linear algebra, projections, least squares
parametric models	$f(x) = f(x;\theta)$, $\theta \in \Theta$	likelihood theory, asymptotics, Fisher information
RKHS (kernel) models	$f \in \mathcal{H}_K$	Hilbert spaces, kernels, representer theorem
nonparametric classes	Sobolev, Hölder balls	functional analysis, approximation theory

4 Data Structures and Their Implications

4.1 The i.i.d. assumption

$Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} P_0$ (randomized trials, survey responses, coin tosses) gives the LLN, the CLT, $\sqrt{n}$-convergence, and classical confidence intervals, tests and regression inference.

4.2 Beyond i.i.d.

- **Dependence**: with positive autocorrelation the effective sample size is below n; for a stationary AR(1), $\mathrm{Var}(\bar{Z}) \approx \frac{\sigma^2}{n}\frac{1+\rho}{1-\rho}$.
- **High dimension**: $p \gg n$. When $p > n$, $Y = X\beta$ is underdetermined and OLS is not unique; structure such as sparsity $\|\beta\|_0 = s \ll n$ is needed.
- **Curse of dimensionality**: covering $[0,1]^p$ by cubes of side ϵ needs $N(\epsilon) \asymp (1/\epsilon)^p$ cubes, so local methods (Ch. 2) need exponentially many observations.
- **Partial observation**: with survival time T and censoring time C we record $\tilde{T} = \min(T, C)$.
- **Measurement error**: $\tilde{X} = X + \delta$ and $Y = f(X) + \varepsilon$; observed relationships may be distorted (attenuation, Ch. 5).

Derivation: why the curse of dimensionality bites local methods

Let $X \sim \mathrm{Unif}[0,1]^p$. A cube capturing a fraction r of the data has side $e_p(r) = r^{1/p}$; for $r = 0.01$, $p = 10$, $e_p \approx 0.63$, so the neighborhood is not local. For $n = 500$ uniform points in the unit ball with $p = 10$, the median distance from the center to the nearest point is about 0.52, tending to 1 as p grows (Exercise 1.2).

5 Exploratory Data Analysis

5.1 Visualization

Distributions of each variable, relationships, correlation and clusters, projections, temporal and spatial patterns, anomalies, missing data, model diagnostics.

5.2 The geometry of the data cloud

Gaussian density contours are ellipses determined by the covariance matrix.

Aspect	Summary	Classical test
center (location)	mean	z-test, t-test
dispersion (spread)	standard deviation	χ^2-test, F-test
orientation (direction)	correlation	t-test for correlation

Shape: skewness $\mathbb{E}[(X-\mu)^3]/\sigma^3$ and kurtosis $\mathbb{E}[(X-\mu)^4]/\sigma^4$ (3 for the normal).

5.3 Other summaries

Quantiles, the five-number summary $(\min, Q_1, \text{median}, Q_3, \max)$, robust statistics (median, IQR $= Q_3 - Q_1$, MAD $= \text{median}\,|X_i - \text{median}\,X|$), boxplots and density plots. One wild value can move the mean arbitrarily far but not the median beyond the range of the remaining data.

Summary

- Statistical learning estimates $P(Y \mid X)$ or the structure of $P(X)$ from finite, noisy samples.
- Without assumptions the target is not identified; $Y = f(X) + \varepsilon$ with a function class fixes flexibility and tools.
- Dependence, high dimension, censoring and measurement error change what can be learned.

6 Selected exercises

Motivation. Short derivations behind four claims of this chapter.

Exercise 1.1 (Variance of the mean under AR(1) dependence)

Let (Z_t) be stationary with mean μ, variance $\sigma^2 > 0$ and $\text{Corr}(Z_s, Z_t) = \rho^{|s-t|}$, $|\rho| < 1$.
(a) Show that $\text{Var}(\bar{Z}_n) = \frac{\sigma^2}{n}\Big[1 + 2\sum_{h=1}^{n-1}\Big(1 - \frac{h}{n}\Big)\rho^h\Big]$.
(b) Show that $n\text{Var}(\bar{Z}_n) \to \sigma^2 \frac{1+\rho}{1-\rho}$.
(c) Interpret $n_{\text{eff}} = n(1-\rho)/(1+\rho)$.

Solution [Supplement].

(a) $\text{Var}(\sum_t Z_t) = \sigma^2 \sum_{s,t} \rho^{|s-t|}$; n pairs have $|s-t| = 0$ and $2(n-h)$ have $|s-t| = h$, $1 \le h \le n-1$. Hence

$$\text{Var}\Big(\sum_t Z_t\Big) = \sigma^2\Big[n + 2\sum_{h=1}^{n-1}(n-h)\rho^h\Big];$$

divide by n^2.

(b) From (a),

$$n\text{Var}(\bar{Z}_n) = \sigma^2\Big[1 + 2\sum_{h=1}^{n-1}\rho^h - \frac{2}{n}\sum_{h=1}^{n-1} h\rho^h\Big].$$

The first sum tends to $\rho/(1-\rho)$; since $\sum_h h|\rho|^h < \infty$, the last term is $\le \frac{2}{n}\sum_{h\ge 1} h|\rho|^h \to 0$. The limit is $\sigma^2\{1 + 2\rho/(1-\rho)\} = \sigma^2(1+\rho)/(1-\rho)$.

(c) $\text{Var}(\bar{Z}_n) \approx \sigma^2/n_{\text{eff}}$: the sample is as informative as n_{eff} independent observations. $\rho = 0.5$ gives $n_{\text{eff}} = n/3$; for $\rho < 0$ the mean is more precise than under independence. ■

Exercise 1.2 (The nearest neighbor is far away)

Let $X_1, \ldots, X_n$ be i.i.d. uniform on $B_p = \{x \in \mathbb{R}^p : \|x\| \le 1\}$ and $D = \min_i \|X_i\|$.
(a) Show that the median of D is $d(p,n) = \big(1 - 2^{-1/n}\big)^{1/p}$.
(b) Evaluate $d(10, 500)$ and show that $d(p,n) \to 1$ as $p \to \infty$ for fixed n.

Solution [Supplement].

(a)

Step 1. $\mathrm{vol}(rB_p) = r^p \,\mathrm{vol}(B_p)$, so $\Pr(\|X_1\| \le r) = r^p$, $0 \le r \le 1$.

Step 2. $\Pr(D > r) = \prod_i \Pr(\|X_i\| > r) = (1 - r^p)^n$, continuous and strictly decreasing from 1 to 0 on $[0, 1]$. The median solves $(1 - m^p)^n = 1/2$, i.e. $m^p = 1 - 2^{-1/n}$.

(b) $d(10, 500) \approx 0.518$. For fixed n, $c = 1 - 2^{-1/n} \in (0, 1)$ and $c^{1/p} = e^{(\log c)/p} \to 1$: in high dimension the "nearest" neighbor is not near. ■

Exercise 1.3 (The median cannot be carried away)

Let $n = 2m + 1$ and $x_1, \ldots, x_n \in \mathbb{R}$. Replace any $k \le m$ observations by arbitrary values; let U be the $n - k$ untouched values, $a = \min U$, $b = \max U$.
(a) Show that the median of the new sample lies in $[a, b]$.
(b) Show that with $m + 1$ replaced values the median can be made arbitrarily large, whereas one replaced value suffices for the mean. Conclude that the breakdown points are $(n + 1)/(2n) \to 1/2$ for the median and $1/n$ for the mean.

Solution [Supplement].

(a) Let $y_{(1)} \le \cdots \le y_{(n)}$ be the new sample, median $y_{(m+1)}$. At most $k \le m$ values are $< a$; if $y_{(m+1)} < a$, the $m + 1$ values $y_{(1)}, \ldots, y_{(m+1)}$ would all be $< a$. Likewise at most m values are $> b$; if $y_{(m+1)} > b$, the $m + 1$ values $y_{(m+1)}, \ldots, y_{(n)}$ would all be $> b$.

(b) Set $m + 1$ values equal to $T > \max_i x_i$; then $y_{(m+1)} = T$, arbitrarily large. Replacing x_1 by $x_1 + t$ moves $\bar{x}$ by t/n, unbounded in t. By (a) no fraction $k/n \le m/n$ breaks the median, so its breakdown point is $(m + 1)/n = (n + 1)/(2n)$. ■

Exercise 1.4 (Least squares when $p > n$)

Let $X \in \mathbb{R}^{n \times p}$ with $p > n$ and $y \in \mathbb{R}^n$.
(a) Show that $X^\top X$ is singular and that the normal equations $X^\top X\beta = X^\top y$ have infinitely many solutions.
(b) Show that all solutions give the same fitted vector $X\hat{\beta}$; if $\mathrm{rank} X = n$, this vector is y itself (zero training error).
(c) If $\mathrm{rank} X = n$, show that $\beta^\dagger = X^\top (XX^\top)^{-1} y$ is the unique solution of minimum Euclidean norm.

Solution [Supplement].

(a) $\mathrm{rank}(X^\top X) = \mathrm{rank} X \le n < p$, so $X^\top X$ is singular and $\dim \mathcal{N}(X) \ge p - n \ge 1$. Since $\mathrm{col}(X^\top X) \subseteq \mathrm{col}(X^\top)$ with equal dimensions, $X^\top y \in \mathrm{col}(X^\top X)$: the system is consistent. If $\hat{\beta}$ solves it, so does $\hat{\beta} + v$ for every $v \in \mathcal{N}(X)$.

(b) For solutions β_1, β_2, $\|X(\beta_1 - \beta_2)\|^2 = (\beta_1 - \beta_2)^\top X^\top X(\beta_1 - \beta_2) = 0$, so $X\beta_1 = X\beta_2$. If $\mathrm{rank} X = n$, $\mathrm{col}(X) = \mathbb{R}^n$, so some β has $X\beta = y$; it solves the normal equations, hence every solution has fitted vector y.

(c) $XX^\top$ is invertible and $X\beta^\dagger = y$; the solution set of $X\beta = y$ is $\beta^\dagger + \mathcal{N}(X)$. As $\beta^\dagger = X^\top w \in \mathcal{N}(X)^\perp$, $\|\beta^\dagger + v\|^2 = \|\beta^\dagger\|^2 + \|v\|^2$ for $v \in \mathcal{N}(X)$, minimized only at $v = 0$. ■